

CSCI 160 Homework 2 — Answer Key

1. Demonstrate the validity of the identity

$$x' + yz = (x' + y)(x' + z).$$

x	y	z	x'	yz	$x' + yz$	$x' + y$	$x' + z$	$(x' + y)(x' + z)$
0	0	0	1	0	1	1	1	1
0	0	1	1	0	1	1	1	1
0	1	0	1	0	1	1	1	1
0	1	1	1	1	1	1	1	1
1	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	1	0
1	1	0	0	0	0	1	0	0
1	1	1	0	1	1	1	1	1

The two final expression columns match for every row.

Answer: $x' + yz = (x' + y)(x' + z)$ is valid.

2. Simplify each expression to a minimum number of literals.

(a)

$$\begin{aligned}
 A'B' + A'B + AB &= A'(B' + B) + AB && \text{factor } A' \\
 &= A' \cdot 1 + AB && \text{complement} \\
 &= A' + AB && \text{identity} \\
 &= (A' + A)(A' + B) && \text{distributive} \\
 &= 1(A' + B) && \text{complement} \\
 &= A' + B. && \text{identity}
 \end{aligned}$$

Answer: $A' + B$

(b)

$$\begin{aligned}
 (x + y)(x + y') &= x + yy' && \text{using } (x + y)(x + z) = x + yz \\
 &= x + 0 && \text{complement} \\
 &= x. && \text{identity}
 \end{aligned}$$

Answer: x

(c)

$$\begin{aligned}
 wx'y + wx'y' + wx &= wx'(y + y') + wx && \text{factor } wx' \\
 &= wx' \cdot 1 + wx && \text{complement} \\
 &= wx' + wx && \text{identity} \\
 &= w(x' + x) && \text{factor } w \\
 &= w \cdot 1 && \text{complement} \\
 &= w. && \text{identity}
 \end{aligned}$$

Answer: w

3. Determine whether the Boolean equation is true or false:

$$xy + y'z + xz = xy + y'z.$$

Do not use a truth table. Start with the extra term xz :

$$\begin{aligned} xz &= xz \cdot 1 \\ &= xz(y + y') \quad \text{complement} \\ &= xyz + xy'z. \quad \text{distributive} \end{aligned}$$

Now substitute this into the left side:

$$\begin{aligned} xy + y'z + xz &= xy + y'z + xyz + xy'z \\ &= (xy + xyz) + (y'z + xy'z) \quad \text{group terms} \\ &= xy(1 + z) + y'z(1 + x) \quad \text{factor} \\ &= xy \cdot 1 + y'z \cdot 1 \quad \text{Theorem: } 1 + x = 1 \\ &= xy + y'z. \quad \text{identity} \end{aligned}$$

Answer: True.

4. Find the complement of

$$F = w'x + yz'.$$

Using DeMorgan's Theorem:

$$\begin{aligned} F' &= (w'x + yz')' \\ &= (w'x)'(yz')' \quad \text{outer OR becomes AND} \\ &= (w + x')(y' + z). \quad \text{flip variables and operators} \end{aligned}$$

So

$$\text{Answer: } F' = (w + x')(y' + z).$$

Now show $F \cdot F' = 0$ and $F + F' = 1$ using the complement postulate on the entire function. Treat the whole expression $w'x + yz'$ as one Boolean expression.

$$\begin{aligned} F \cdot F' &= (w'x + yz')((w'x + yz')') \\ &= 0 \quad \text{Postulate 5: } E \cdot E' = 0. \end{aligned}$$

Since $(w'x + yz')' = (w + x')(y' + z)$, this also shows

$$(w'x + yz')(w + x')(y' + z) = 0.$$

$$\begin{aligned} F + F' &= (w'x + yz') + ((w'x + yz')') \\ &= 1 \quad \text{Postulate 5: } E + E' = 1. \end{aligned}$$

Since $(w'x + yz')' = (w + x')(y' + z)$, this also shows

$$(w'x + yz') + (w + x')(y' + z) = 1.$$

5. Obtain the truth table of

$$F(x, y, z) = x'y + xz.$$

x	y	z	$x'y$	xz	F
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	1
0	1	1	1	0	1
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	0	0	0
1	1	1	0	1	1

The rows where $F = 1$ are 2, 3, 5, 7.

$$\begin{aligned} F &= x'yz' + x'yz + xy'z + xyz \\ &= \sum m(2, 3, 5, 7). \end{aligned}$$

The rows where $F = 0$ are 0, 1, 4, 6.

$$\begin{aligned} F &= (x + y + z)(x + y + z')(x' + y + z)(x' + y' + z) \\ &= \prod M(0, 1, 4, 6). \end{aligned}$$

6. Convert each expression into both sum-of-products form and product-of-sums form.

(a)

$$(A' + B)(B + C)$$

SOP:

$$\begin{aligned} (A' + B)(B + C) &= A'B + A'C + BB + BC && \text{distributive} \\ &= A'B + A'C + B + BC && \text{Theorem 1: } BB = B \\ &= B + A'C. && \text{Theorem 6: absorption} \end{aligned}$$

POS:

$$\text{Answer: } (A' + B)(B + C)$$

$$\text{SOP answer: } B + A'C, \quad \text{POS answer: } (A' + B)(B + C).$$

(b)

$$w + x(y + z')$$

SOP:

$$w + x(y + z') = w + xy + xz'.$$

POS:

$$w + x(y + z') = (w + x)(w + y + z') \quad \text{using } a + bc = (a + b)(a + c).$$

$$\text{SOP answer: } w + xy + xz', \quad \text{POS answer: } (w + x)(w + y + z').$$

7. Bitwise operations.

$$A = 11010110, \quad B = 10011010.$$

Bitwise XOR:

$$\begin{array}{r} 11010110 \\ \oplus 10011010 \\ \hline 01001100 \end{array}$$

For bitwise NOR, first take the bitwise OR, then complement it:

$$\begin{array}{r} 11010110 \\ + 10011010 \\ \hline 11011110 \end{array} \implies (11011110)' = 00100001.$$

$$\text{Answer: } A \oplus B = 01001100, \quad \text{NOR}(A, B) = 00100001.$$

8. Draw the logic diagram for

$$F = AB' + A'B + BC.$$

Use one NOT gate to make A' , one NOT gate to make B' , three AND gates to make AB' , $A'B$, and BC , and one OR gate to combine them.

